

Many Treatments, Some Instruments: Sharp Identification and Uniform Inference in Treatment Selection Models

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Nonparametric IV with Many Treatments and/or Instruments

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→ **Allow:** assumptions on treatment selection equation,
e.g., shape restrictions on g_D and/or dist. restrictions on V

→ **Avoid:** assumptions on outcome equation, i.e. on g_Y or ε_Y

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Definition. (Treatment Selection Model, TSM)

A **treatment selection model** is a 3-tuple $\mathbf{S} = (\mathcal{V}, \mathcal{F}_{\mathcal{V}}, \mathcal{G}_D)$, where

$\mathcal{V}$ is a set, $\mathcal{F}_{\mathcal{V}}$ is a set of distributions over $\mathcal{V}$,
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$Q \in \mathcal{M}$ satisfies $\mathbf{S}$ iff there exists $(Y, D, Z, \mathbf{V}, \varepsilon_Y, g_D, g_Y)$ satisfying (IV-1)-(IV-3) such that:

- (i) $\text{supp}(\mathbf{V}) \subseteq \mathcal{V}_{\mathbf{S}}$, $F_{\mathbf{V}} \in \mathcal{F}_{\mathcal{V}}$, and $g_D \in \mathcal{G}_D$,
- (ii) it generates Q via (PO-1)-(PO-2).

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E.g., $\mathcal{V} := \mathbb{R}$, $\mathcal{F}_V$ unrestricted, $\mathcal{G}_D := \{(z, v) \mapsto \mathbb{1}\{v \leq \nu(z)\} | \nu \text{ increasing}\}$

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- **Potential Outcome Means/Treatment Effects:** For known coeffs. α_d and known functions κ_d , means of the form

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- **Ratios of Outcome Means:** For known $\alpha_d, \beta_d, \kappa_d$ (some $\beta_d \neq 0$):

$$\frac{\mathbb{E} \left[\sum_{d \in \mathcal{D}} \alpha_d \kappa_d(Y_d) \mid V \in \mathcal{V}^* \right]}{\mathbb{E} \left[\sum_{d \in \mathcal{D}} \beta_d \kappa_d(Y_d) \mid V \in \mathcal{V}^* \right]}.$$

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→ e.g., in settings with multiple dimensions of treatment, $\frac{\mathbb{E}[Y_{(0,1)} - Y_{(0,0)}]}{\mathbb{E}[Y_{(1,1)} - Y_{(0,0)}]}$

Examples of Potential Applications

- Multi-dimensional instrument(s) or treatment(s)
 - **Multi-dimensional treatment**: disentangling different dimensions
 - Mediation: want to separate direct, indirect, and total effects
 - More generally: settings with complex incentive schemes, or complicated relationship between Z and D
- Ordered Treatments: e.g., Impact of Incarceration, Schooling etc.
- Nonparametric Discrete Choice Models
 - **MTO, Head Start**, College Choice...
- Other settings with multi-dimensional heterogeneity
 - Survey/RCT Non-Response Bias (e.g., Dutz et al., 2022)
- Exclusion Restriction Violations and/or Sample Selection Problems
 - See, e.g., Kroft, Mourifié, and Vayalinal (2024)
- Sensitivity analysis for point-identified models
 - In the spirit of Noack (2021)

Related IV Literature: Identification

Point Identification: Angrist and Imbens (1995)... Heckman and Vytlacil (1999, 2001, 2007a,b)... Kirkebøen et al. (2016); Heckman and Pinto (2018); Lee and Salanié (2018); Mountjoy (2022)...

Partial Identification: Manski (1989, 1990); Balke and Pearl (1997)... Beresteanu et al. (2012)... Mogstad et al. (2018); Torgovitsky (2019); Kitagawa (2021); Kamat et al. (2023); Marx (2024); Bai et al. (2024)...

- **This paper:** sharp under full IV independence for any selection model; allows multivalued treatments and continuous outcomes; tractable, but not “black box”
- **This paper cannot:** incorporate general assumptions on outcomes, or continuous instruments

Specification Testing : Kitagawa (2015); Mourifié and Wan (2017); Kédagni and Mourifié (2020); Sun (2023)...

- **This paper:** sharp test for general selection models with discrete IV

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For inclusion test: parameter of interest may not be a feature of Q !

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→ In fact, S has identifying power *only* through restrictions on π

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→ $\mathcal{R}_\pi$ is a feature of $\mathbf{S}$, does not depend on data P

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→ now, $\pi = \pi_Q$ is a feature of Q

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$$\begin{aligned}\mathcal{R}_\pi &:= \left\{ \left(\mathbb{P}_F \left[(D^*(V; g), \mathbb{1}\{V \in \mathcal{V}^*\}) = \tau \right] : \tau \in \mathcal{T} \right) \mid g \in \mathcal{G}_D, F \in \mathcal{F}_V \right\} \\ &= \{ \pi_Q \mid Q \in \mathcal{M} \text{ and } Q \text{ satisfies } \mathbf{S} \},\end{aligned}$$

where $D^*(v; g) := (g(z, v) : z \in \mathcal{Z})$.

→ $\mathcal{R}_\pi$ is a feature of $\mathbf{S}$, does not depend on data P

Theorem 1.

For any treatment selection model $\mathbf{S}$, $\mathcal{Q}_\mathbf{S} = \{Q \in \mathcal{Q} \mid \pi_Q \in \mathcal{R}_\pi(\mathbf{S})\}$.

→ $\mathbf{S}$ has identifying power on features of Q **only** through dist. of T

→ $\mathbf{S}$ is **sharply** summarized by $(\mathcal{T}, \mathcal{R}_\pi)$

IV Mixture Model

For any (y, d, z) , we have

$$\underbrace{\mathbb{P}(Y \leq y, D = d, Z = z)}_{\text{observed data}} = \sum_{\tau \in \mathcal{T}} \underbrace{\mathbb{1}\{\tau(z) = d\}}_{\text{known from model}} \underbrace{\mathbb{P}(Y_d \leq y, \mathbf{T} = \tau, Z = z)}_{\text{unobserved}},$$

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or, using densities and dividing both sides by $\mathbb{P}(Z = z)$,

$$\underbrace{\mathbb{P}(D = d | Z = z) f_{Y|D=d, Z=z}(y)}_{=: z\text{-th entry of vector } \mathbf{p}\mathbf{f}_d(y)} = \sum_{\tau \in \mathcal{T}} \underbrace{\mathbb{1}\{\tau(z) = d\}}_{=: (z, \tau)\text{-th entry of matrix } \mathbf{R}_d} \underbrace{\mathbb{P}(\mathbf{T} = \tau) f_{Y_d|\mathbf{T}=\tau}(y)}_{=: \tau\text{-th entry of vector } \boldsymbol{\xi}_d(y)}$$

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For each $d \in \mathcal{D}$ and $y \in \mathcal{Y}$, get **linear system** $(\star)(d, y)$:

$$\mathbf{R}_d \boldsymbol{\xi}_d(y) = \mathbf{pf}_d(y) \quad \text{and} \quad \boldsymbol{\xi}_d(y) \geq 0 \quad (\star)$$

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→ When $(\star)$ is underdetermined, $\boldsymbol{\xi}_d$ is generally not point-identified.

→ Note: $\int_{\mathcal{Y}} \boldsymbol{\xi}_d(y) dy = \boldsymbol{\pi}$

Identified Set for π

Since $\int_{\mathcal{Y}} \xi_d(y) dy = \pi$, and ξ_d satisfies $(\star)$, we must have

$$\pi \in \left\{ \int_{\mathcal{Y}} \mathbf{x}(y) dy \mid \begin{array}{l} \text{integrable } \mathbf{x} : \mathcal{Y} \rightarrow \mathbb{R}_{\geq 0}^{\mathcal{T}} \\ \mathbf{R}_d \mathbf{x}(y) = \mathbf{p} \mathbf{f}_d(y), \forall y \in \mathcal{Y} \end{array} \right\} =: \Pi_d$$

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→ This must hold for all d , so $\pi \in \bigcap_{d \in \mathcal{D}} \Pi_d =: \Pi_{\mathcal{T}}$

→ To satisfy S, also need $\pi \in \mathcal{R}_{\pi}$

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Theorem 2 (Sharp Testable Implications of Selection Models)

For any treatment selection model S,

$$\Pi_S := \{ \pi_Q \mid Q \in \mathcal{Q}_S \} = \mathcal{R}_{\pi}(S) \cap \Pi_{\mathcal{T}(S)}$$

In particular, $\mathcal{Q}_S \neq \emptyset$ if and only if $\mathcal{R}_{\pi}(S) \cap \Pi_{\mathcal{T}(S)} \neq \emptyset$.

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→ **Goal:** Tractable representation of $\mathcal{R}_{\pi} \cap \Pi_{\mathcal{T}}$

Candidates for π : Support Function

Every non-empty closed convex set $C \subseteq \mathbb{R}^n$ is uniquely determined by its **support function** $h_C : \mathbb{R}^n \rightarrow \mathbb{R}$, defined as

$$h_C(\mathbf{c}) := \sup_{\mathbf{x} \in C} \mathbf{c}^\top \mathbf{x} .$$

Define the shorthand $h_d(\mathbf{c}; \mathbf{p}\mathbf{f}_d(y)) := \sup_{\mathbf{x} \geq 0 : \mathbf{R}_d \mathbf{x} = \mathbf{p}\mathbf{f}_d(y)} \mathbf{c}^\top \mathbf{x} .$

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Lemma 1

Suppose $\mathcal{T}$ and P are such that for all $y \in \mathcal{Y}$, there exists $\mathbf{x} \geq 0$ with $\mathbf{R}_d \mathbf{x} = \mathbf{p}\mathbf{f}_d(y)$. Then, Π_d is a non-empty convex set such that, for all $\mathbf{c} \in \mathbb{R}^{\mathcal{T}}$

$$h_{\Pi_d}(\mathbf{c}) = \int_{\mathcal{Y}} h_d(\mathbf{c}; \mathbf{p}\mathbf{f}_d(y)) \, dy .$$

Further, Π_d is a convex polytope.

Candidates for π : Moment Representation

Lemma 2

Suppose for all $y \in \mathcal{Y}$, there exists $x \geq 0$ with $\mathbf{R}_d x = \mathbf{pf}_d(y)$. Then, for $c : \mathcal{Y} \rightarrow \mathbb{R}^T$ with $c_{(\tau)}(\cdot) = 0$ for all “ d -never-takers”, there exists $m_d(c(\cdot), \mathbf{pf}_d(\cdot))$ such that, for all $y \in \mathcal{Y}$,

$$m_d(c(y); \mathbf{pf}_d(y)) \in \underset{\substack{\lambda: \|\lambda\|_\infty \leq M \|c(y)\|_\infty, \\ \mathbf{R}_d^\top \lambda \geq c(y)}}{\operatorname{argmin}} \lambda^\top \mathbf{pf}_d(y),$$

with M a constant (does not depend on P or c , only $\mathbf{R}_d$)

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with M a constant (does not depend on P or c , only $\mathbf{R}_d$), and

$$\begin{aligned} & \int_{\mathcal{Y}} h_d(c(y), \mathbf{p}\mathbf{f}_d(y)) dy \\ &= \sum_{z \in \mathcal{Z}} \mathbb{E} \left[m_{d,(z)}(c(Y); \mathbf{p}\mathbf{f}_d(Y)) \mathbb{1}\{D = d\} \mid Z = z \right] \\ &= \mathbb{E} \left[\sum_{z \in \mathcal{Z}} \frac{\mathbb{1}\{D = d, Z = z\}}{\mathbb{P}(Z = z)} m_{d,(z)}(c(Y); \mathbf{p}\mathbf{f}_d(Y)) \right]. \end{aligned}$$

Candidates for π : Moment Representation

For any conformable matrix B , with rows $\{b_1, \dots, b_k\}$, define shorthands

$$\int_y \mathbf{h}_d(B; \mathbf{p}\mathbf{f}_d(y)) dy := \begin{bmatrix} \int_y \mathbf{h}_d(b_1; \mathbf{p}\mathbf{f}_d(y)) dy \\ \vdots \\ \int_y \mathbf{h}_d(b_k; \mathbf{p}\mathbf{f}_d(y)) dy \end{bmatrix},$$
$$\mathbb{E}[\mathbf{m}_d(B; Y, D, Z)] := \begin{bmatrix} \mathbb{E} \left[\sum_{z \in \mathcal{Z}} \frac{\mathbb{1}\{D=d, Z=z\}}{\mathbb{P}(Z=z)} \mathbf{m}_d(b_1; \mathbf{p}\mathbf{f}_d(Y))_{(z)} \right] \\ \vdots \\ \mathbb{E} \left[\sum_{z \in \mathcal{Z}} \frac{\mathbb{1}\{D=d, Z=z\}}{\mathbb{P}(Z=z)} \mathbf{m}_d(b_k; \mathbf{p}\mathbf{f}_d(Y))_{(z)} \right] \end{bmatrix}.$$

Candidates for π : Moment Representation

Corollary 1

There exists a **known matrix** $\mathbf{A}_d$, depending only on $\mathbf{R}_d$, such that, if for all $y \in \mathcal{Y}$, there exists $x \geq 0$ with $\mathbf{R}_d x = \mathbf{p}\mathbf{f}_d(y)$, then

$$\begin{aligned}\Pi_d &= \left\{ \mathbf{x} \in \mathbb{R}^T \mid \mathbf{A}_d \mathbf{x} \leq \int_{\mathcal{Y}} \mathbf{h}_d(\mathbf{A}_d; \mathbf{p}\mathbf{f}_d(y)) dy \right\} \\ &= \{ \mathbf{x} \in \mathbb{R}^T \mid \mathbf{A}_d \mathbf{x} \leq \mathbb{E}[\mathbf{m}_d(\mathbf{A}_d; Y, D, Z)] \} .\end{aligned}$$

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Define $\mathbf{A}_{\mathcal{T}} := \begin{bmatrix} \mathbf{A}_1 \\ \vdots \\ \mathbf{A}_{N_D} \end{bmatrix}$. If the condition holds for all $d \in \mathcal{D}$, then,

$$\Pi_{\mathcal{T}} = \left\{ \mathbf{x} \in \mathbb{R}^{\mathcal{T}} \mid \mathbf{A}_{\mathcal{T}} \mathbf{x} \leq \begin{bmatrix} \mathbb{E}[\mathbf{m}_1(\mathbf{A}_1; Y, D, Z)] \\ \vdots \\ \mathbb{E}[\mathbf{m}_{N_D}(\mathbf{A}_{N_D}; Y, D, Z)] \end{bmatrix} \right\} ,$$

and $\Pi_{\mathcal{T}}$ is a **compact convex polytope**.

Candidates for π : Moment Representation

Corollary 1

There exists a **known matrix** $\mathbf{A}_d$, depending only on $\mathbf{R}_d$, such that, if for all $y \in \mathcal{Y}$, there exists $x \geq 0$ with $\mathbf{R}_d x = \mathbf{p} \mathbf{f}_d(y)$, then

$$\begin{aligned}\Pi_d &= \left\{ x \in \mathbb{R}^{\mathcal{T}} \mid \mathbf{A}_d x \leq \int_{\mathcal{Y}} \mathbf{h}_d(\mathbf{A}_d; \mathbf{p} \mathbf{f}_d(y)) dy \right\} \\ &= \{ x \in \mathbb{R}^{\mathcal{T}} \mid \mathbf{A}_d x \leq \mathbb{E}[\mathbf{m}_d(\mathbf{A}_d; Y, D, Z)] \} .\end{aligned}$$

Define $\mathbf{A}_{\mathcal{T}} := \begin{bmatrix} \mathbf{A}_1 \\ \vdots \\ \mathbf{A}_{N_D} \end{bmatrix}$. If the condition holds for all $d \in \mathcal{D}$, then,

$$\Pi_{\mathcal{T}} = \left\{ x \in \mathbb{R}^{\mathcal{T}} \mid \mathbf{A}_{\mathcal{T}} x \leq \begin{bmatrix} \mathbb{E}[\mathbf{m}_1(\mathbf{A}_1; Y, D, Z)] \\ \vdots \\ \mathbb{E}[\mathbf{m}_{N_D}(\mathbf{A}_{N_D}; Y, D, Z)] \end{bmatrix} \right\} ,$$

and $\Pi_{\mathcal{T}}$ is a **compact convex polytope**.

$\rightarrow \Pi_{\mathcal{T}} \neq \emptyset$ iff above **finite linear system** has a solution

Sharp Specification Test for Treatment Selection Models

$$\text{Define } \mathbb{E}[\mathbf{m}_{\mathcal{T}}(Y, D, Z)] := \begin{bmatrix} \mathbb{E}[\mathbf{m}_1(\mathbf{A}_1; Y, D, Z)] \\ \vdots \\ \mathbb{E}[\mathbf{m}_{N_D}(\mathbf{A}_{N_D}; Y, D, Z)] \end{bmatrix}.$$

Theorem 3 (Specification Test for Convex $\mathcal{R}_{\pi}$)

Suppose $\mathcal{R}_{\pi}(\mathbf{S})$ is a closed convex set and, for all $y \in \mathcal{Y}$, there exists $x \geq 0$ with $\mathbf{R}_d x = \mathbf{p}f_d(y)$. Then, for all $\mathbf{c} \in \mathbb{R}^{\mathcal{T}}$,

$$\begin{aligned} h_{\Pi_S}(\mathbf{c}) &= \inf_{\mathbf{u} \in \mathbb{R}^{\mathcal{T}}} \{h_{\mathcal{R}_{\pi}}(\mathbf{c} - \mathbf{u}) + h_{\Pi_{\mathcal{T}}}(\mathbf{u})\} \\ &= \inf_{\boldsymbol{\lambda} \geq 0} \{h_{\mathcal{R}_{\pi}}(\mathbf{c} - \mathbf{A}_{\mathcal{T}}^{\top} \boldsymbol{\lambda}) + \boldsymbol{\lambda}^{\top} \mathbb{E}[\mathbf{m}_{\mathcal{T}}(Y, D, Z)]\}. \end{aligned}$$

Further,

$$\mathcal{Q}_S \neq \emptyset \iff \inf_{\boldsymbol{\lambda} \geq 0: \|\mathbf{A}_{\mathcal{T}}^{\top} \boldsymbol{\lambda}\|_1 = 1} \{h_{\mathcal{R}_{\pi}}(-\mathbf{A}_{\mathcal{T}}^{\top} \boldsymbol{\lambda}) + \boldsymbol{\lambda}^{\top} \mathbb{E}[\mathbf{m}_{\mathcal{T}}(Y, D, Z)]\} \geq 0.$$

Sharp Specification Test for Treatment Selection Models

When restrictions implied by $\mathcal{R}_\pi$ are linear, $\mathcal{R}_\pi$ is a convex polytope:

$$\mathcal{R}_\pi = \{\pi \in [0, 1]^T \mid \mathbf{A}_\mathcal{R} \pi \leq \mathbf{a}_\mathcal{R}\}.$$

Sharp Specification Test for Treatment Selection Models

When restrictions implied by $\mathcal{R}_\pi$ are **linear**, $\mathcal{R}_\pi$ is a **convex polytope**:

$$\mathcal{R}_\pi = \{\pi \in [0, 1]^\mathcal{T} \mid \mathbf{A}_\mathcal{R} \pi \leq \mathbf{a}_\mathcal{R}\}.$$

Let $W := (Y, D, Z)$, $\mathbf{A}_\mathcal{S} := \begin{bmatrix} \mathbf{A}_\mathcal{T} \\ \mathbf{A}_\mathcal{R} \end{bmatrix}$ and $\mathbb{E}[m_\mathcal{S}(W)] := \begin{bmatrix} \mathbb{E}[m_\mathcal{T}(W)] \\ \mathbf{a}_\mathcal{R} \end{bmatrix}$.

Theorem 4 (Specification Test for Linear $\mathcal{R}_\pi$)

Suppose $\mathcal{R}_\pi(\mathcal{S})$ is a **closed convex polytope** as above and, for all $y \in \mathcal{Y}$, there exists $x \geq 0$ with $\mathbf{R}_d x = \mathbf{p}f_d(y)$. Then, for all $c \in \mathbb{R}^\mathcal{T}$,

$$h_{\Pi_\mathcal{S}}(c) = \inf_{\lambda \geq 0: \mathbf{A}_\mathcal{S}^\top \lambda = c} \lambda^\top \mathbb{E}[m_\mathcal{S}(W)].$$

Further, $\mathcal{Q}_\mathcal{S} \neq \emptyset \iff \inf_{\substack{\lambda \geq 0: \|\lambda\|_1 = 1, \\ \mathbf{A}_\mathcal{S}^\top \lambda = 0}} \lambda^\top \mathbb{E}[m_\mathcal{S}(W)] \geq 0.$

Sharp Specification Test, II

When matrices $\mathbf{A}_d$ not available, can use following alt. formulation.

Theorem 4B

Suppose $\mathcal{R}_\pi(\mathbf{S})$ is a closed convex polytope and, for all $y \in \mathcal{Y}$, there exists $\mathbf{x} \geq 0$ with $\mathbf{R}_d \mathbf{x} = \mathbf{p} \mathbf{f}_d(y)$. Then, for all $\mathbf{c} \in \mathbb{R}^T$,

$$h_{\Pi_S}(\mathbf{c}) = \inf_{\substack{(\mathbf{w}_d:d): \\ \mathbf{w}_d \in \mathbb{R}^T}} \left\{ \mathbf{a}_{\mathcal{R}}^\top \left(\mathbf{c} - \sum_d \mathbf{w}_d \right) + \sum_{d \in \mathcal{D}} \mathbb{E}[\mathbf{m}_d(\mathbf{w}_d; W)] \right\}$$

Further,

$$\mathcal{Q}_S \neq \emptyset \iff \inf_{\substack{(\mathbf{w}_d:d): \mathbf{w}_d \in \mathbb{R}^T, \\ \|\mathbf{w}_d:d\|_\infty \leq 1}} \left\{ \mathbf{a}_{\mathcal{R}}^\top \left(\sum_d \mathbf{w}_d \right) + \sum_{d \in \mathcal{D}} \mathbb{E}[\mathbf{m}_d(\mathbf{w}_d; W)] \right\} \geq 0.$$

Counterfactual Outcome Means and Treatment Effects

For simplicity, consider $\theta := \mathbb{E} [\kappa(Y_d) \mid \mathbf{V} \in \mathcal{V}^*]$, for some $\kappa : \mathcal{Y} \rightarrow \mathbb{R}$.
 $\rightarrow \theta_Q$ well-defined whenever $\mathbb{P}_Q(\mathbf{V} \in \mathcal{V}^*) > 0$.

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→ θ_Q well-defined whenever $\mathbb{P}_Q(\mathbf{V} \in \mathcal{V}^*) > 0$.

→ Define identified set for θ under $\mathbf{S}$:

$$\Theta_{\mathbf{S}} := \begin{cases} \emptyset & \text{iff } \mathcal{Q}_{\mathbf{S}} = \emptyset, \\ \Theta_0 & \text{if } \inf_{Q \in \mathcal{Q}_{\mathbf{S}}} \mathbb{P}_Q(\mathbf{V} \in \mathcal{V}^*) = 0, \\ \left\{ \frac{\mathbb{E}_Q[\kappa(Y_d) \mathbb{1}\{\mathbf{V} \in \mathcal{V}^*\}]}{\mathbb{P}_Q(\mathbf{V} \in \mathcal{V}^*)} \mid Q \in \mathcal{Q}_{\mathbf{S}} \right\} & \text{otherwise,} \end{cases}$$

where Θ_0 is a known set, typically $\mathcal{Y}$.

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where Θ_0 is a known set, typically $\mathcal{Y}$.

Recall $\xi_d := (\mathbb{P}(T = \tau) f_{Y_d|T=\tau} : \tau \in \mathcal{T})$, and so,

$$\int_{\mathcal{Y}} \kappa(y) \xi_d(y) dy = (\mathbb{E}[\kappa(Y_d) \mathbb{1}\{T = \tau\}] : \tau \in \mathcal{T}).$$

Counterfactual Outcome Means and Treatment Effects

Note that $\mathbb{1}\{V \in \mathcal{V}^*\} = \sum_{\tau \in \mathcal{T}} \mathbb{1}\{T(V) = \tau\} \mathbb{1}\{T^{-1}(\tau) \subseteq \mathcal{V}^*\}$.

Counterfactual Outcome Means and Treatment Effects

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$$\int_{\mathcal{Y}} c^\top \xi_d(y) dy = \mathbb{P}(V \in \mathcal{V}^*) \quad \text{and} \quad \int_{\mathcal{Y}} c^\top \kappa(y) \xi_d(y) dy = \mathbb{E}[\kappa(Y_d) \mathbb{1}\{V \in \mathcal{V}^*\}].$$

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$$\rightarrow \theta = \frac{\mathbb{E}[\kappa(Y_d) \mathbb{1}\{V \in \mathcal{V}^*\}]}{\mathbb{P}(V \in \mathcal{V}^*)} = \frac{c^\top (\mathbb{E}[\kappa(Y_d) \mathbb{1}\{T = \tau\}] : \tau \in \mathcal{T})}{c^\top \pi}.$$

Counterfactual Outcome Means and Treatment Effects

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Define $c := (\mathbb{1}\{T^{-1}(\tau) \subseteq \mathcal{V}^*\} : \tau \in \mathcal{T})$ so that

$$\int_{\mathcal{Y}} c^\top \xi_d(y) dy = \mathbb{P}(V \in \mathcal{V}^*) \quad \text{and} \quad \int_{\mathcal{Y}} c^\top \kappa(y) \xi_d(y) dy = \mathbb{E}[\kappa(Y_d) \mathbb{1}\{V \in \mathcal{V}^*\}].$$

$$\rightarrow \theta = \frac{\mathbb{E}[\kappa(Y_d) \mathbb{1}\{V \in \mathcal{V}^*\}]}{\mathbb{P}(V \in \mathcal{V}^*)} = \frac{c^\top (\mathbb{E}[\kappa(Y_d) \mathbb{1}\{T = \tau\}] : \tau \in \mathcal{T})}{c^\top \pi}.$$

Therefore, $\Theta_S = \{\gamma \in \Theta_0 \mid \exists (m, p) \in \Upsilon_d : p \in \Pi_S, c^\top m = \gamma c^\top p\}$

Counterfactual Outcome Means and Treatment Effects

Note that $\mathbb{1}\{V \in \mathcal{V}^*\} = \sum_{\tau \in \mathcal{T}} \mathbb{1}\{T(V) = \tau\} \mathbb{1}\{T^{-1}(\tau) \subseteq \mathcal{V}^*\}$.

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Therefore, $\Theta_S = \{\gamma \in \Theta_0 \mid \exists (m, p) \in \Upsilon_d : p \in \Pi_S, c^\top m = \gamma c^\top p\}$, where

$$\Upsilon_d := \left\{ \left(\int_{\mathcal{Y}} \kappa(y) x(y) dy, \int_{\mathcal{Y}} x(y) dy \right) \mid \begin{array}{l} \text{integrable } x : \mathcal{Y} \rightarrow \mathbb{R}_{\geq 0}^{\mathcal{T}} \\ \mathbf{R}_d x(y) = \mathbf{p} \mathbf{f}_d(y) \quad \forall y \in \mathcal{Y} \end{array} \right\}.$$

Counterfactual Outcome Means and Treatment Effects

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$\rightarrow \Upsilon_d$ always convex, generally **not a polytope** unless $\kappa(\mathcal{Y})$ finite

Counterfactual Outcome Means and Treatment Effects

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$$\rightarrow \theta = \frac{\mathbb{E}[\kappa(Y_d) \mathbb{1}\{V \in \mathcal{V}^*\}]}{\mathbb{P}(V \in \mathcal{V}^*)} = \frac{c^\top (\mathbb{E}[\kappa(Y_d) \mathbb{1}\{T = \tau\}] : \tau \in \mathcal{T})}{c^\top \pi}.$$

Therefore, $\Theta_S = \{\gamma \in \Theta_0 \mid \exists (m, p) \in \Upsilon_d : p \in \Pi_S, c^\top m = \gamma c^\top p\}$, where

$$\Upsilon_d := \left\{ \left(\int_{\mathcal{Y}} \kappa(y) x(y) dy, \int_{\mathcal{Y}} x(y) dy \right) \mid \begin{array}{l} \text{integrable } x : \mathcal{Y} \rightarrow \mathbb{R}_{\geq 0}^{\mathcal{T}} \\ \mathbf{R}_d x(y) = \mathbf{p} \mathbf{f}_d(y) \quad \forall y \in \mathcal{Y} \end{array} \right\}.$$

$\rightarrow \Upsilon_d$ always convex, generally **not a polytope** unless $\kappa(\mathcal{Y})$ finite

$\rightarrow$ **Goal:** Tractable test for whether $\{(m, p) \in \Upsilon_d \mid p \in \Pi_S, c^\top m = \gamma c^\top p\} = \emptyset$

Inclusion Test for Mean Outcomes

Lemma 3 (Support Function of Υ_d)

Suppose for all $y \in \mathcal{Y}$, there exists $x \geq 0$ with $\mathbf{R}_d x = \mathbf{p}\mathbf{f}_d(y)$. Then, Υ_d is a non-empty convex set such that, for any $(c_1, c_2) \in \mathbb{R}^T \times \mathbb{R}^T$,

$$h_{\Upsilon_d}(c_1, c_2) = \int_{\mathcal{Y}} h_d(\kappa(y) c_1 + c_2; \mathbf{p}\mathbf{f}_d(y)) dy .$$

Inclusion Test for Mean Outcomes

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Theorem 5 (Inclusion Test for Outcome Means)

If $\mathcal{R}_\pi$ is closed and convex, and for all $y \in \mathcal{Y}$, there exists $x \geq 0$ with $\mathbf{R}_d x = \mathbf{p}\mathbf{f}_d(y)$, then, whenever $c_{(\tau)} = 0$ for “ d -never-takers”,

$$\Theta_S = \left\{ \gamma \in \Theta_0 \left| \inf_{\substack{(\mathbf{u}, t) \in \mathbb{R}^T \times \mathbb{R} \\ \|\mathbf{u}, t\|_\infty \leq 1}} \left(\int_{\mathcal{Y}} h_d(t(\kappa(y) - \gamma) \mathbf{c} + \mathbf{u}; \mathbf{p}\mathbf{f}_d(y)) dy + h_{\Pi_S}(-\mathbf{u}) \right) \geq 0 \right. \right\}$$

Inclusion Test for Mean Outcomes: Moment Representation

When $\mathcal{R}_\pi = \{\pi \mid \mathbf{A}_\mathcal{R} \pi \leq \mathbf{a}_\mathcal{R}\}$, with $W := (Y, D, Z)$:

$$\Theta_S = \left\{ \gamma \in \Theta_0 \mid \inf_{\substack{(t, \boldsymbol{\lambda}): \boldsymbol{\lambda} \geq 0, \\ \|(t, \boldsymbol{\lambda})\|_\infty \leq 1}} \underbrace{\left\{ \mathbb{E} [m_d(t(\kappa(Y) - \gamma) \mathbf{c} - \mathbf{A}_S^\top \boldsymbol{\lambda}; W)] + \boldsymbol{\lambda}^\top \mathbb{E} [m_S(W)] \right\}}_{=: \Psi_\gamma(t, \boldsymbol{\lambda})} \geq 0 \right\}$$

Inclusion Test for Mean Outcomes: Moment Representation

When $\mathcal{R}_\pi = \{\pi \mid \mathbf{A}_\mathcal{R} \pi \leq \mathbf{a}_\mathcal{R}\}$, with $W := (Y, D, Z)$:

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Let $\underline{\Psi}_\gamma$ be infimum of unconditional moment inequalities problem above.

→ Can construct confidence region for θ by inverting, over γ , test of

$$H_0 : \underline{\Psi}_\gamma \geq 0 \quad \text{vs.} \quad H_1 : \underline{\Psi}_\gamma < 0$$

→ Ψ_γ convex in $(t, \boldsymbol{\lambda}) \implies$ finite-dimensional convex program

→ $\mathbf{m}_d$ functions depend on P and must be estimated

Computation of m_d and h_d : Setup

Given vector (or matrix) x and a subset B of coordinates of x , let:

- x_B be the subvector of x corresponding to the coordinates in B
- x_{B^c} be the subvector consisting of the remaining coordinates

“Basic solution” to (primal) LP $\max_{x \geq 0: \mathbf{R}_d x = b} c^\top x$ is of the form

$$x_B = \mathbf{R}_{d,B}^{-1} b \text{ and } x_{B^c} = 0 ,$$

where B is a subset of columns of $\mathbf{R}_d$ such that $\mathbf{R}_{d,B}$ is invertible.

Basic solution optimal iff both of the following hold,

1. It is primal-feasible: $\mathbf{R}_{d,B}^{-1} b \geq 0$,
2. It is dual-feasible: $c_B^\top \mathbf{R}_{d,B}^{-1} \mathbf{R}_{d,B^c} - c_{B^c}^\top \geq 0$.

→ **Key:** Solve LP at one $y' \in \mathcal{Y} \implies$ get sol'ns for y -values in region of $\mathcal{Y}$

Let $P_d(B) := \{b \mid \mathbf{R}_{d,B}^{-1} b \geq 0\}$, $D_d(B) := \{c \mid c_B^\top \mathbf{R}_{d,B}^{-1} \mathbf{R}_{d,B^c} - c_{B^c}^\top \geq 0\}$.

Computation of m_d and h_d : Algorithm

Algorithm 1

Input Functions $c : \mathcal{Y} \rightarrow \mathbb{R}^T$ and $\mathbf{pf}_d : \mathcal{Y} \rightarrow \mathbb{R}^Z$,

Solver $\text{LP}_d(c, b)$, returns an **optimal basis** for $\max_{x \geq 0: \mathbf{R}_d x = b} c^\top x$.

Output Functions $h_d : \mathcal{Y} \rightarrow \mathbb{R}$ and $m_d : \mathcal{Y} \rightarrow \mathbb{R}^Z$.

-
- 1: Set $\mathcal{Y}^\dagger := \emptyset$ and $\mathcal{G} := \emptyset$.
 - 2: **While** $\mathcal{Y}^\dagger \neq \mathcal{Y}$:
 - 3: Choose arbitrary $\tilde{y} \in \mathcal{Y} \setminus \mathcal{Y}^\dagger$ and update $\mathcal{G} \leftarrow \mathcal{G} \cup \{\tilde{y}\}$,
 - 4: Set $B(\tilde{y}) = \text{LP}_d(c(y), \mathbf{pf}_d(y))$,
 - 5: Set $O(\tilde{y}) := \{y \in \mathcal{Y} \mid \mathbf{pf}_d(y) \in P_d(B(\tilde{y})), c(y) \in D_d(B(\tilde{y}))\} \setminus \mathcal{Y}^\dagger$,
 - 6: **Update** $\mathcal{Y}^\dagger \leftarrow \mathcal{Y}^\dagger \sqcup O(\tilde{y})$.
 - 7: **Return** $h_d(y) = \sum_{\tilde{y} \in \mathcal{G}} \mathbb{1}\{y \in O(\tilde{y})\} c(y)^\top \mathbf{R}_{d, B(\tilde{y})}^{-1} \mathbf{pf}_d(y)$,
 $m_d(y) = \sum_{\tilde{y} \in \mathcal{G}} \mathbb{1}\{y \in O(\tilde{y})\} \mathbf{R}_{d, B(\tilde{y})}^{-\top} c(y)$.

Estimation and Inference: Setup

For a class of data distributions $\mathbf{P}$, consider the testing problems

(1) Specification Test: $H_0 : P \in \mathbf{P}_S$ vs. $H_1 : P \in \mathbf{P} \setminus \mathbf{P}_S$, where

$$\mathbf{P}_S := \{P \in \mathbf{P} \mid \mathcal{Q}_S(P) \neq \emptyset\} = \left\{ P \in \mathbf{P} \mid \inf_{\substack{\lambda \geq 0: \|\lambda\|_1=1, \\ \mathbf{A}_S^\top \lambda = 0}} \lambda^\top \mathbb{E}_P[\mathbf{m}_S(W)] \geq 0 \right\}.$$

(2) Inclusion Test, for $\gamma \in \Theta_0$: $H_0 : P \in \mathbf{P}_S(\gamma)$ vs. $H_1 : P \in \mathbf{P} \setminus \mathbf{P}_S(\gamma)$, where

$$\mathbf{P}_S(\gamma) := \{P \in \mathbf{P} \mid \gamma \in \Theta_S(P)\} = \left\{ P \in \mathbf{P} \mid \inf_{\substack{(t, \lambda): \lambda \geq 0, \\ \|(t, \lambda)\|_\infty \leq 1}} \Psi_{\gamma, P}(t, \lambda) \geq 0 \right\}.$$

→ **Goal:** Uniform size control over broad class $\mathbf{P}$, pointwise exactness at every fixed boundary null.

Specification Test: 3-Fold Sample Splitting Approach

Target Estimand: $(\lambda^*)^\top \mathbb{E}[m_S(W)]$, where λ^* solves the problem in (1).

Split sample into 3 folds: $\left\{ \left(Y_i^{(k)}, D_i^{(k)}, Z_i^{(k)} \right) \right\}_{i=1}^{n_k}$ for $k = 1, 2, 3$.

Denote by $\hat{\mathbb{E}}_n^{(k)}$ sample mean over k -th fold: $\hat{\mathbb{E}}_n^{(k)}[X] = \frac{1}{n_k} \sum_{i=1}^{n_k} X_i^{(k)}$.

1. Estimate m_d on fold 1, so that for $k > 1$:

$$\hat{\mathbb{E}}_n^{(k)} \left[\hat{m}_d^{(1)}(c; W) \right] := \hat{\mathbb{E}}_n^{(k)} \left[\sum_{z \in \mathcal{Z}} \frac{\mathbb{1}\{D = d, Z = z\}}{\hat{\mathbb{P}}_n^{(k)}(Z = z)} m_d \left(c; \hat{\mathbf{p}}_d^{(1)}(Y) \right)_{(z)} \right].$$

2. Estimate λ^* on fold 2 using $\hat{\lambda}_S^{(2)}$ that solves LP:

$$\hat{\lambda}_S^{(2)} \in \underset{\substack{\lambda \geq 0: \|\lambda\|_1 = 1, \\ \mathbf{A}_S^\top \lambda = 0}}{\operatorname{argmin}} \lambda^\top \hat{\mathbb{E}}_n^{(2)} \left[\hat{m}_S^{(1)}(W) \right].$$

3. Estimate $(\lambda^*)^\top \mathbb{E}[m_S(W)]$ on fold 3 using $\left(\hat{\lambda}_S^{(2)} \right)^\top \hat{\mathbb{E}}_n^{(3)} \left[\hat{m}_S^{(1)}(W) \right]$

The Specification Test

Let $\hat{\Sigma}_n^{(3)}$ be a suitable fold-3 covar. estimator, use **studentized** test stat.:

$$\hat{T}_n^{ST} := \sqrt{n_3} \frac{(\hat{\lambda}_S^{(2)})^\top \hat{\mathbb{E}}_n^{(3)} [\hat{m}_S^{(1)}(W)]}{\sqrt{(\hat{\lambda}_S^{(2)})^\top \hat{\Sigma}_n^{(3)} \hat{\lambda}_S^{(2)}}}$$

For testing (1) at level α , with $z_\alpha := \Phi^{-1}(\alpha)$, use rule:

$$\text{Reject } H_0 \text{ iff } \hat{T}_n^{ST} < z_\alpha . \quad (\text{ST})$$

Why this works:

- Under $P \in \mathbf{P}_0$: cond'l on folds 1-2, $\hat{T}_n^{ST} \rightsquigarrow N(0, 1)$
- By iterated expectations, also holds unconditionally
- $(\hat{\lambda}_S^{(2)})^\top \mathbb{E}_P [\hat{m}_S^{(1)}(W)] \geq (\lambda_P^*)^\top \mathbb{E}_P [m_S(W)] \implies \text{conservative test}$

Validity of Specification Test

Precisely,

$$\begin{aligned} & \left(\hat{\lambda}_S^{(2)} \right)^\top \hat{\mathbb{E}}_n^{(3)} \left[\hat{m}_S^{(1)}(W) \right] - \left(\lambda_P^* \right)^\top \mathbb{E}_P[m_S(W)] \\ &= \left(\hat{\lambda}_S^{(2)} \right)^\top \hat{\mathbb{E}}_n^{(3)} \left[\hat{m}_S^{(1)}(W) \right] - \left(\hat{\lambda}_S^{(2)} \right)^\top \mathbb{E}_P \left[\hat{m}_S^{(1)}(W) \right] \\ & \quad + \underbrace{\left(\hat{\lambda}_S^{(2)} \right)^\top \mathbb{E}_P \left[\hat{m}_S^{(1)}(W) \right] - \left(\hat{\lambda}_S^{(2)} \right)^\top \mathbb{E}_P[m_S(W)]}_{\geq 0} \\ & \quad + \underbrace{\left(\hat{\lambda}_S^{(2)} \right)^\top \mathbb{E}_P[m_S(W)] - \left(\lambda_P^* \right)^\top \mathbb{E}_P[m_S(W)]}_{\geq 0} \\ & \geq \left(\hat{\lambda}_S^{(2)} \right)^\top \hat{\mathbb{E}}_n^{(3)} \left[\hat{m}_S^{(1)}(W) \right] - \left(\hat{\lambda}_S^{(2)} \right)^\top \mathbb{E}_P \left[\hat{m}_S^{(1)}(W) \right] \end{aligned}$$

→ Using $\left(\hat{\lambda}_S^{(2)} \right)^\top \mathbb{E}_P \left[\hat{m}_S^{(1)}(W) \right]$ for $\left(\lambda_P^* \right)^\top \mathbb{E}_P[m_S(W)]$ can't hurt validity.

Uniform Validity of Specification Test

Throughout the remainder, maintain the following assumption on $\mathbf{P}$.

Assumption P.1

For all $z \in \mathcal{Z}$, $\inf_{P \in \mathbf{P}} \mathbb{P}_P(Z = z) > 0$.

We now have the following result for the uniform validity of test (ST).

Theorem 6

Suppose $\mathcal{R}_\pi$ is a convex polytope and $\mathcal{T}$ satisfies Condition **F**. Then, under Assumption **P.1** and a mild uniform non-degeneracy condition on the fold-3 variance term,

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathbf{P}_S} \mathbb{P}_P(\hat{T}_n^{ST} < z_\alpha) \leq \alpha$$

Pointwise Exactness of Spec. Test: “Anti-Concentration” Req.

For exactness, require following assumption on P , for suitable $\delta \geq 0$:

Assumption AC(δ) (“Anti-Concentration of Linear Combinations”)

For $d \in \mathcal{D}$ and $\mathbf{u} \in \mathbb{R}^T$,

$$\limsup_{t \downarrow 0} \frac{1}{t^\delta} \mathbb{E}_P \left[\frac{\mathbb{1} \{0 < \mathbf{u}^\top \mathbf{p}\mathbf{f}_{d,P}(Y) \leq t\}}{f_{Y,P}(Y)} \right] < \infty$$

Requires control of frequency of “near-ties” in relevant directions $\mathbf{u}$

- Required control is only for each $\mathbf{u}$, not uniform in $\mathbf{u}$
- Exact ties OK, can’t concentrate mass where two bases *almost* tie
- Assumption **AC**(0) follows from tail control, automatic if Y bounded

Pointwise Exactness of Spec. Test

Theorem 7

Suppose $\mathcal{R}_\pi$ is a convex polytope and $\mathcal{T}$ satisfies Condition **F**. Let $\hat{\mathbf{p}}f_d$ be an estimator such that, for some deterministic $r_n \uparrow \infty$,

$$\sup_{y \in \mathcal{Y}} \|\mathbf{p}f_{d,P}(y) - \hat{\mathbf{p}}f_{d;n}(y)\|_\infty = O_P(r_n^{-1}) ,$$

for $d \in \mathcal{D}$. Suppose P satisfies Assumption **AC**(δ). If either

- (i) $r_n = \sqrt{n}$ and $\delta \geq 0$, or
- (ii) $r_n \asymp n^\alpha$ (upto logs) and $\delta > \max\{0, (1/2\alpha) - 1\}$, then

$$\lim_{n \rightarrow \infty} \mathbb{P}_P \left(\hat{T}_n^{ST} < z_\alpha \right) = \alpha$$

→ Continuous scalar Y , twice-differentiable densities: need $\delta > 1/4$

Inclusion Test: 2-Fold Sample Splitting Approach i

Split sample into 2 folds: $\left\{ \left(Y_i^{(k)}, D_i^{(k)}, Z_i^{(k)} \right) \right\}_{i=1}^{n_k}$ for $k = 1, 2$.

Goal: Estimator for $\underline{\Psi}_{\gamma}^{(2)} := \inf_{(t, \lambda) \in \Lambda} \Psi_{\gamma, n}^{(2)}(t, \lambda)$ & suitable crit. value

1. Est. $\hat{\underline{\Psi}}_n^{(2)}$ on fold-2 via iterated cuts & fold-1 moment f'ns $\hat{m}_d^{(1)}$:

- Given $w_0 := (t_0, \lambda_0)$, compute $\hat{\Psi}_n^{(2)}(w_0)$ and subgradient $\hat{g}^{(2)}(w_0)$
- By convexity, can construct a “cut” to get affine lower bound:

$$\ell_{w_0}(w) := \hat{\Psi}_n^{(2)}(w_0) + \hat{g}^{(2)}(w_0)^\top (w - w_0) \leq \hat{\Psi}_n^{(2)}(w).$$

- Once we have $K > 0$ such cuts, solve LP defined by cuts:

$$\min_{(w, r): w \in \Lambda} r : r \geq \ell_{w_j}(w), j = 0, \dots, K$$

to get optimal direction w_K and a lower estimate $\hat{\Psi}_{n, K}^{(2)}(w_K) \leq \hat{\underline{\Psi}}_n^{(2)}$.

- Evaluate $\hat{\Psi}_n^{(2)}$ at w_K , get upper bound: $\hat{\Psi}_{n, K}^{(2)}(w_K) \leq \hat{\underline{\Psi}}_n^{(2)} \leq \hat{\Psi}_n^{(2)}(w_K)$
- Add cuts until approx. err. small at $\sqrt{n}$ -scale $\rightarrow$ use $\hat{\Psi}_n^{(2)}(w_K)$ as est. $\hat{\underline{\Psi}}_n^{(2)}$

Inclusion Test: 2-Fold Sample Splitting Approach ii

2. Cond'l on fold-1,

$$\sqrt{n_2} \left(\hat{\underline{\Psi}}_n^{(2)} - \underline{\Psi}_{P; m^{(1)}} \right) \rightsquigarrow \inf_{\mathbf{w} \in \underline{\Lambda}_P} \left(\hat{\mathbb{E}}_n^{(2)} \left[\mathbf{m}_{\gamma, \mathbf{w}}^{(1)}(W) \right] - \mathbb{E}_P \left[\mathbf{m}_{\gamma, \mathbf{w}}^{(1)}(W) \right] \right)$$

where, taking fold-1 moment functions as true:

- $\underline{\Psi}_{P; m^{(1)}}$ is infimum at the true P , $\underline{\Lambda}_P$ is the associated argmin set,
- $\mathbf{m}_{\gamma, \mathbf{w}}^{(1)}$ is combined fold-1 moment functions in objective function of (2)

→ **Challenge:** The optimization problem in limit law is not convex!

3. Estimate an outer estimate for argmin set using cuts:

$$\hat{\Lambda}_n := \left\{ \mathbf{w} \in \Lambda \mid \hat{\Psi}_{n, K}^{(2)}(\mathbf{w}_K) \leq \hat{\underline{\Psi}}_n^{(2)} + \tau_n \right\}$$

4. Use exchangeable/multiplier bootstrap, solve suitable perturbed LPs over $\hat{\Lambda}_n$

Uniform Integrability Condition for Inclusion Test

Assumption P.2

For all $d \in \mathcal{D}$ and $z \in \mathcal{Z}$,

$$\lim_{\lambda \rightarrow \infty} \sup_{P \in \mathbf{P}} \mathbb{E}_P \left[\kappa(Y)^2 \mathbb{1}_{\{|\kappa(Y)| > \lambda\}} \mid D = d, Z = z \right] = 0 \text{ and}$$

$$\lim_{\lambda \rightarrow \infty} \sup_{P \in \mathbf{P}} \mathbb{E}_P \left[\left(\frac{\kappa(Y) - \eta_P}{\sigma_P} \right)^2 \mathbb{1}_{\left\{ \left| \frac{\kappa(Y) - \eta_P}{\sigma_P} \right| > \lambda \right\}} \mid D = d, Z = z \right] = 0,$$

where $\eta_P := \mathbb{E}_P[\kappa(Y) \mid D = d, Z = z]$ and $\sigma_P := \mathbb{E}_P[(\kappa(Y) - \eta_P)^2]^{1/2}$.

Inclusion test

binary Z	Design	Inner cov.	Outer cov.	Reject at γ_0	Median length	Sec./CR
	Weak full	1.00	1.00	0.00	2.594	24.3
	Strong monotone bdry	0.98	1.00	0.02	0.453	28.1
	Weak monotone bdry	0.96	0.97	0.04	1.625	28.8

Spec test

$Z = (Z_1, Z_2)$, partial monotonicity. Z_k binary.

Method	Reps	Rejection Rate	Sec./rep
Known A_d	100	0.05	0.57
IC, 5 cuts,	100	0.04	4.91
FSST, $K = 10$	100	0.00	2.86
FSST, $K = 20$	100	0.02	9.61

Seconds per rep for finer discretization:

FSST, $K = 50$: 39.25

FSST, $K = 100$: 238.70

Thank you!!

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